

# A further conjecture on OEIS A165381, settled by the proof already in hand

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## Abstract

OEIS A165381 carries more than one conjecture. One of them has a proof; this paper settles a DIFFERENT one, and it needs no new model and no new engine – only the linear recurrence already established for the sequence. A claim about a C-finite sequence whose own characteristic polynomial is a multiple of the established one is decided by polynomial divisibility, and where the claim is an identity rather than a recurrence, by divisibility together with a finite window of agreement.

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## 1 The premise

The entry's first terms are 37, 285, 1012, 2467, 4708, 7748, 11588, 16228, ..., with offset 2. The sequence satisfies

$$a(n) = 3a(n-1) - 3a(n-2) + 1a(n-3),$$

a linear recurrence of order 3 with characteristic polynomial

$$q(x) = x^3 - 3x^2 + 3x - 1.$$

This recurrence is proved in this project; it is the premise of everything below and nothing else is assumed. It was confirmed against every term the entry publishes from the index where it begins to hold, with at least  $3 + 2$  further terms after that index.

## 2 The further conjecture

Separately from the conjecture already settled, the entry records

$$a(n) = 4*(817 - 540*n + 100*n^2) \text{ for } n > 5.$$

The further claim is a CLOSED FORM, and for that the divisibility is not the whole argument. It gives that  $a$  and the closed form  $f$  satisfy the same linear recurrence; it does not say they are the same solution of it. Write  $p$  for the annihilator of  $f$ ,

$$p(x) = x^3 - 3x^2 + 3x - 1,$$

of order  $D = 3$ . Since  $q \mid p$  and  $a$  satisfies the recurrence of  $q$ ,  $a$  satisfies that of  $p$  as well;  $f$  satisfies it by construction. Two solutions of a linear recurrence of order  $D$  with nonzero leading coefficient agree everywhere as soon as they agree at  $D$  consecutive indices, and  $a$  and  $f$  were checked to agree at  $n = 9, \dots, 11$ , which lies inside the terms the entry publishes and past the index from which  $a$  provably satisfies  $p$ . Hence  $a(n) = f(n)$  for every  $n$  in the claimed range.

**Theorem 1.** *The conjecture displayed above is true.*

### 3 What is and is not claimed here

This is a second result on an entry that already has one, and it is a second result only because it is a DIFFERENT statement. A generating function whose denominator's reciprocal is exactly  $q$  says precisely what the proved recurrence says, in another notation; such restatements are identified and are not counted. The conjecture settled here is not of that kind.